Perishable Lot-Sizing with Period-Dependent Shelf Lives

Compact Mixed-Integer Programming Model (Freshest-First)

Indices and Sets

- $T = \{1, 2, \ldots, \mathbf{T}\}$: set of planning periods.
- $I = \{1, 2, \ldots, \mathbf{I}\}$: set of items.
- $t \in T$: production period, $u \in T$: consumption period.

Perishability

Shelf life. Units of i produced at t can be used at $u \geq t$ only if

$$u - t \leq m_{it}.$$

Arc set. Variables X_{itu} exist only when $t \leq u \leq \min\{\, \mathbf{T},\, t + m_{it} - 1 \,\}$.

Parameters

d_{iu}	demand of item i in period u.
c_{it}	unit production cost of item i in period t.
h_{it}	unit holding cost of item i per period of age.
s_{it}	setup cost of item i in period t.
κ_t	total production capacity in period t.
cap_{it}	(we said only capt but if we want we can keep cap across items constant anyway so I kept this notation) per-item capacity in period t.
m_{it}	shelf life (periods) for production (i, t).
W	(if given) warehouse capacity (end-of-period inventory).
μ_{it}	tight big-M for (i, t): $\mu_{it} = \sum_{u=t}^{\min\{\mathbf{T},\, t+m_{it}-1\}} d_{iu}$.
g_{itu}	generalized unit cost: $g_{itu} = c_{it} + h_{it}(u - t)$.

Decision Variables

$X_{itu} \geq 0$	quantity of i produced at t and consumed at u (only if $t \leq u \leq \min\{\mathbf{T},\, t+m_{it}-1\}$).
$Y_{it} \in \{0, 1\}$	setup decision for item i in period t.
$S_{iu\tau} \geq 0$	**freshest-first slack** for item i, demand period u, threshold τ

Objective

$$\min \sum_{i=1}^{\mathbf{I}} \sum_{t=1}^{\mathbf{T}} \sum_{u=t,\ldots,\min\{\mathbf{T},\, t+m_{it}-1\}} g_{itu}\, X_{itu} \;+\; \sum_{i=1}^{\mathbf{I}} \sum_{t=1}^{\mathbf{T}} s_{it}\, Y_{it} \tag{1}$$

Constraints

Global capacity (per production period). For all $t = 1, \ldots, \mathbf{T}$,

$$\sum_{i=1}^{\mathbf{I}} \sum_{u=t,\ldots,\min\{\mathbf{T},\, t+m_{it}-1\}} X_{itu} \;\le\; \kappa_t. \tag{2a}$$

Capacity across items is constant (only time matters). For all $i = 1, \ldots, I$, $t = 1, \ldots, T$,

$$\sum_{i=1}^{I} x_{it} \;\le\; \text{cap}_t$$

Setup linking (big-M tied to (i,t)). For all it,

$$\sum_{u=t,\ldots,\min\{\mathbf{T},\, t+m_{it}-1\}} X_{itu} \;\le\; \mu_{it}\, Y_{it} \tag{2b}$$

Warehouse capacity (if given). For all $u = 1, \ldots, \mathbf{T}-1$,

$$\sum_{i=1}^{\mathbf{I}} \sum_{t=1}^{u} \sum_{w=\max\{u+1,t\},\ldots,\min\{\mathbf{T},\, t+m_{it}-1\}} X_{itw} \;\le\; W \tag{2c}$$

Demand balance (no backorders). For all $i = 1, \ldots, \mathbf{I}$, $u = 1, \ldots, \mathbf{T}$

$$\sum_{\substack{t=1,\ldots,u \\ t\le u\le \min\{\mathbf{T},\, t+m_{it}-1\}}} X_{itu} \;=\; d_{iu} \tag{2d}$$

Freshest-first (LEFO). For all $i = 1, \ldots, \mathbf{I}$, $u = 1, \ldots, \mathbf{T}$, and thresholds $\tau = 1, \ldots, u-1$:

$$S_{iu\tau} \;\ge\; d_{iu} \;-\; \sum_{\substack{t=\tau+1,\ldots,u \\ t\le u\le \min\{\mathbf{T},\, t+m_{it}-1\}}} X_{itu} \tag{2e}$$

$$\sum_{\substack{t=1,\ldots,\tau \\ t\le u\le \min\{\mathbf{T},\, t+m_{it}-1\}}} X_{itu} \;\le\; S_{iu\tau} \tag{2f}$$

Meaning: Older cohorts ($t \le \tau$) are used only to cover the residual after allocating from newer cohorts ($t \ge \tau{+}1$) towards d_{iu}.

Domains.

$$X_{itu} \ge 0, \quad Y_{it} \in \{0,1\}, \quad S_{iu\tau} \ge 0, \tag{2g}$$